

Class 02 Assessment — How Machines Learn

Bong Study Hub — Foundation Batch 2026 · Class 02: How Machines Learn

Estimated time: 25–35 minutes **Total points:** 20 **Tools needed:** pen and paper, or a simple document — nothing else

Instructions

- Complete this **independently**, in **your own words**.
 - These scenarios are new — you haven't seen them before. That's intentional. You're not being asked to recall a sentence from class; you're being asked to **reason using the same ideas** in a fresh situation.
 - **No coding, no math, and no advanced AI knowledge** is required or expected.
 - There is no single "correct sentence" for most of these questions — if your reasoning is sound and shows real understanding, it counts, even if it doesn't match the exact wording from class.
 - If you're unsure, **explain your thinking anyway** rather than leaving a question blank.
 - Work through the questions in order — several scenarios build on each other.
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Question 1 — The Rule Problem (3 points)

A college wants to build a simple system that automatically checks ID-card photo submissions and decides whether each photo is "acceptable" — for example, no sunglasses, a plain background, facing the camera — **using only rules a person writes by hand**.

As more students submit photos, in more lighting conditions, poses, and styles, why might this hand-written-rules approach become difficult to manage?

Explain the role of variety, exceptions, and changing situations in your answer.

Question 2 — The Pivot (2 points)

Continuing the ID-photo scenario: the college realizes that hand-written rules are becoming impractical.

What could they give the computer instead of more rules, and what would the computer eventually build and use from that?

Question 3 — Pipeline Reasoning (4 points)

A campus food-delivery app is shown many past orders, each already labeled "**delivered on time**" or "**delivered late**." Later, a brand-new order comes in.

Using this scenario, explain what DATA, LEARNING, MODEL, and PREDICTION would each mean here.

Write one or two sentences for each — and show how the four stages connect to one another, not just four separate definitions.

Question 4 — Traditional Programming vs. Machine Learning (3 points)

A college is deciding how to automatically sort photos submitted to its event page into "**Indoor**" or "**Outdoor**."

Approach A: A developer sits down and writes a list of explicit conditions describing what counts as "Indoor" versus "Outdoor."

Approach B: The system is given many examples of photos already labeled "Indoor" or "Outdoor," and uses a learning process to build a model that can make a prediction about new, unlabeled photos.

- a. Which approach is traditional programming?
- b. Which approach is Machine Learning?
- c. In your own words, what is the key difference between the two approaches?

Question 5 — Prediction Is Not Certainty (3 points)

A campus event-recommendation app suggests a workshop to a student, but the student ends up not finding it interesting at all.

Does this automatically mean the app wasn't using Machine Learning? Explain why or why not.

Question 6 — Data Quality / An Unseen Case (3 points)

A campus system is learning to guess whether a parked vehicle outside a building is "**a student's ride**" or "**a delivery vehicle.**" During learning, it only ever saw two kinds of examples: small bicycles (labeled "student's ride") and large delivery vans (labeled "delivery vehicle"). One day, it encounters a food-delivery worker's **motorbike** parked outside.

What might happen, and why?

Connect your answer to what the system did or didn't see while learning.

Question 7 — An Unfamiliar Application (2 points)

Imagine a campus app that tries to predict **which floor of the library is likely to have free seats in the next 30 minutes.** (This is a made-up example — you're not being asked about any real app.)

1. What kind of examples/data might this system use?
2. What kind of useful pattern might matter?
3. What kind of prediction/output could it produce?

Optional Reflection — Not Scored

This section is entirely optional and will not affect your score.

Is there anything about Class 02's ideas you'd still like to understand better? This just helps your instructor know what's worth revisiting.

Before You Submit — Self-Check

- ☐ I answered all 7 questions in my own words.
- ☐ I explained my reasoning, even where I wasn't fully sure.
- ☐ I did not use code or programming syntax anywhere.
- ☐ I reviewed my work before submitting.